KCE PATHWAYS · GRADE 3 · UNIT 4: PARTS AND PIECES · WEEK 19 · MATH

Morning at the Balloon Fiesta — Teacher Lesson Plan & Answer Key

Learning objectives: Students tell and write time to the nearest minute, find elapsed time and start or end times on a number line, solve one- and two-step problems with liquid volume in liters and mass in grams and kilograms, and choose sensible units and estimates for real objects.

Duration: 35–45 minutes
Materials: Computer or tablet with the lesson open · pencil · printed Pre & Post sheet

Lesson flow (as it appears on screen)

1. Title and name: the student types first and last name so results go to KingCretot Experience. Read the Pathway Points rules together: 2 tries per question, points only for a correct first try, and the answer and reason appear after a second miss. Introduce Balloon Fiesta week in Albuquerque, New Mexico, with pilot Ms. Yazzie and her ground crew, Ivy, Nico and Luz.
2. Warm-Up: Comparing Fractions Review (10 pts, Week 18 skill): a typed comparison and a which-explanation-is-correct item. If a student misses both, review "same denominator: compare numerators; same numerator: compare piece size."
3. Picture It: Morning at the Balloon Fiesta at Balloon Fiesta Park, with the sun rising over the Sandia Mountains. Ask: what jobs on a launch morning need a clock, a jug, or a scale?
4. Word Wrangler: 6 vocab cards (elapsed time, minute hand, a.m. and p.m., liter, gram, kilogram).
5. Pencil Break (Pre): the student opens or prints the Pre/Post sheet (grade3/prepost/KCE_Math_PrePost_G3W19.pdf) and completes Page 1 (Pre) before any teaching. Do not help with answers yet; it shows what they already know.
6. Telling Time to the Minute (10 pts): reading a clock to the minute and a find-the-mistake item. Tip: count by fives to the last big number, then count on by ones; the hour is the number the hour hand has passed.
7. Elapsed Time: How Much Time Passes (10 pts): finding an end time and a two-step start-time problem. Tip: draw an open number line and jump to the next hour first.
8. Liquid Volume: Liters of Water and Cocoa (10 pts): reading a measuring jug and two-step liter problems (multiply, then add or subtract).
9. Mass: Grams and Kilograms (10 pts): reading a scale in grams and a find-the-mistake estimate item. Tip: a paper clip is about 1 gram; a big bag of flour is about 1 kilogram.
10. Launch Field Challenge: Two-Step Problems (10 pts): mixed time and volume problems. Ask the student to name step 1 before solving.
11. Bonus Practice: The Morning Rounds (XP only, not scored).
12. Match-Up Round (10 pts, 2 per first-try match): Set 1 matches vocabulary, Set 2 matches elapsed times and clock times, Set 3 matches volume and mass problems to answers.
13. Fix the Sentence (10 pts, 5 each): typed answers are checked exactly (capital letters, names, end mark, list commas); curly quotes and extra spaces are forgiven. Both serial-comma and no-serial-comma list forms are accepted. Remind the student to keep the same words in the same order.
14. Speed Arena and Keyboard Corral (XP only, not scored).

15. Boss Battle, Part 1: The Big Launch Morning (10 pts, 5 each): an elapsed-time problem that crosses noon and a which-explanation-is-correct item about liters.
16. Boss Battle, Part 2: The Last Landing (10 pts, 5 each): adding masses in grams and a find-the-mistake time item (for example, 10:75 is not a time).
17. Pencil Break (Post): the student turns to Page 2 (Post) of the same sheet (grade3/prepost/KCE_Math_PrePost_G3W19.pdf) and completes it, plus the ★ Challenge if time allows, before seeing the score. The sheet buttons let the student open or print it again.
18. Results (Fiesta Morning Complete!): the score out of 100 is sent to KingCretot Experience. 85 or higher shows “Fiesta Hero!” with the congratulations message “Fiesta hero! You timed and measured the whole launch morning!” and a big confetti burst; below 85 shows “Great Effort Today!” with encouragement (“Every balloon crew gets better with practice.”), and the next attempt brings a fresh question set (sets rotate 1, 2, 3). Both messages remind the student to finish the Post-Lesson Challenge sheet.

Lesson answer key with reasoning

Every scored item the student sees, in order, across all three rotating question sets. Each item allows 2 tries and earns Pathway Points only on a correct first try. The "why" is the same reasoning shown on screen after a second missed try.

WARM-UP: COMPARING FRACTIONS REVIEW (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
w1_s0	Set 1: Type <, >, or = to compare: ⅜ ___ ⅝	**<** (also accepted: is less than, less than, less)	Both fractions are eighths, so the pieces are the same size. 3 pieces is fewer than 5 pieces, so 3/8 < 5/8.
w2_s0	Set 1: Which explanation is correct? Why is ⅓ greater than ⅙? [Both are 1 piece, but sixths are smaller pieces, because the whole is cut into more parts. / 6 is bigger than 3, so 1/6 must be bigger. / Both fractions have a 1 on top, so they are equal. / Thirds are smaller pieces than sixths.]	**Both are 1 piece, but sixths are smaller pieces, because the whole is cut into more parts.**	With the same numerator, the fraction with the smaller denominator has bigger pieces. 1/3 > 1/6.
w1_s1	Set 2: Type <, >, or = to compare: ¾ ___ ⅚	**>** (also accepted: is greater than, greater than, greater, more than)	Both fractions have 2 pieces. Fourths are bigger pieces than sixths, so 2/4 > 2/6.
w2_s1	Set 2: Find the mistake. Ivy says ⅛ is greater than ¼ because 8 is greater than 4. What went wrong? [Eighths are smaller pieces than fourths, so 1/8 is less than 1/4. / Nothing is wrong. 8 is more than 4, so 1/8 is greater. / The two fractions are equal because both have a 1 on top. / She should add: 1/8 + 1/4.]	**Eighths are smaller pieces than fourths, so 1/8 is less than 1/4.**	The bigger the denominator, the more pieces the whole is cut into, so each piece is smaller. 1/8 < 1/4.
w1_s2	Set 3: Type <, >, or = to compare: ⅚ ___ ⅝	**>** (also accepted: is greater than, greater than, greater, more than)	Both fractions have 5 pieces. Sixths are bigger pieces than eighths, so 5/6 > 5/8.

ID	Question	Answer	Why (explain to student)
w2_s2	Set 3: Two breakfast burritos are the same size. Nico ate ¾ of one burrito. Luz ate ⅜ of the other. Who ate more? [Nico, because 3/4 of a burrito is more than 3/8 of the same-size burrito. / Luz, because 8 is greater than 4. / They ate the same amount, because both ate 3 pieces. / You cannot compare fractions of the same-size burrito.]	**Nico, because 3/4 of a burrito is more than 3/8 of the same-size burrito.**	The numerators match, so compare the size of the pieces. Fourths are bigger than eighths, so 3/4 > 3/8.

TELLING TIME TO THE MINUTE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
t1_s0	Set 1: Ivy looks at the crew tent clock. What time does it show? Type it like 7:48.	**8:37** (also accepted: 8:37 a.m., 8:37 p.m., 8:37 am, 8:37 pm, 8:37am, 8:37pm)	The hour hand has passed the 8. The minute hand is 2 marks past the 7: 35 + 2 = 37. It is 8:37.
t2_s0	Set 1: Find the mistake. Nico reads this clock as 5:52. What went wrong? [The hour hand has not reached the 5 yet, so it is still the 4 o'clock hour. The time is 4:52. / Nothing is wrong. The hour hand is closest to the 5, so it is 5:52. / The minute hand shows 10, so it is 5:10. / The time is 10:20, because the minute hand points near the 10.]	**The hour hand has not reached the 5 yet, so it is still the 4 o'clock hour. The time is 4:52.**	The hour hand is between the 4 and the 5, so the hour is 4. The minute hand is 2 marks past the 10: 50 + 2 = 52. It is 4:52.
t1_s1	Set 2: What time does the clock in the crew tent show? Type it like 7:48.	**11:26** (also accepted: 11:26 a.m., 11:26 p.m., 11:26 am, 11:26 pm, 11:26am, 11:26pm)	The hour hand has passed the 11. The minute hand is 1 mark past the 5: 25 + 1 = 26. It is 11:26.
t2_s1	Set 2: No picture this time! The hour hand is between the 6 and the 7. The minute hand points 3 small marks past the 7. What time is it?	**6:38** (also accepted: 6:38 a.m., 6:38 p.m., 6:38 am, 6:38 pm, 6:38am, 6:38pm)	The hour hand is between the 6 and the 7, so the hour is 6. Minute hand: 7 × 5 = 35, then 3 more is 38. It is 6:38.
t1_s2	Set 3: What time does the clock show? Type it like 7:48.	**3:09** (also accepted: 3:09 a.m., 3:09 p.m., 3:09 am, 3:09 pm, 3:09am, 3:09pm)	The hour hand has just passed the 3. The minute hand is 4 marks past the 1: 5 + 4 = 9. It is 3:09.
t2_s2	Set 3: Which explanation is correct? How can you tell this clock shows 9:43? [The hour hand is between the 9 and the 10, and the minute hand is 3 small marks past the 8: 40 + 3 = 43. / The minute hand points at the 4, and the hour hand points at the 3. / Each small mark is 5 minutes, so 8 small marks make 43. / The hour hand points right at the 10, so the hour is 10.]	**The hour hand is between the 9 and the 10, and the minute hand is 3 small marks past the 8: 40 + 3 = 43.**	Count by fives to the 8 (40), then count on by ones (41, 42, 43). The hour hand has passed the 9 but not the 10.

ELAPSED TIME: HOW MUCH TIME PASSES (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
e1_s0	Set 1: A balloon lifts off at 9:25 a.m. and floats for 48 minutes. What time does it land?	**10:13 a.m.** (also accepted: 10:13, 10:13 am, 10:13am, 10:13a.m., 10:13 a.m, 10:13 a. m.)	Jump from 9:25 to 10:00: 35 minutes. 48 − 35 = 13 minutes left. 10:00 + 13 minutes = 10:13 a.m.

ID	Question	Answer	Why (explain to student)
e2_s0	Set 1: Two steps! Luz's pin trading table closed at 2:10 p.m. It was open for 25 minutes. What time did it open?	**1:45 p.m.** (also accepted: 1:45, 1:45 pm, 1:45pm, 1:45p.m., 1:45 p.m, 1:45 p. m.)	Jump back from 2:10 to 2:00: 10 minutes. 25 − 10 = 15 minutes left. 2:00 back 15 minutes is 1:45 p.m.
e1_s1	Set 2: Nico started folding the crew tarps at 1:35 p.m. and finished at 2:20 p.m. How many minutes did he work?	**45 minutes** (also accepted: 45, 45 minute, 45 min, 45 mins, 45min, 45mins)	From 1:35 to 2:00 is 25 minutes. From 2:00 to 2:20 is 20 minutes. 25 + 20 = 45 minutes.
e2_s1	Set 2: Find the mistake. Ivy says the time from 7:50 a.m. to 8:15 a.m. is 65 minutes, because 815 − 750 = 65. What went wrong? [An hour has 60 minutes, not 100. 7:50 to 8:00 is 10 minutes, then 15 more: 25 minutes. / Nothing is wrong. 815 − 750 = 65, so it is 65 minutes. / It is 1 hour 15 minutes, because 8 − 7 = 1. / It is 35 minutes, because 50 − 15 = 35.]	**An hour has 60 minutes, not 100. 7:50 to 8:00 is 10 minutes, then 15 more: 25 minutes.**	Clock times are not regular numbers. Jump to the hour: 7:50 to 8:00 is 10 minutes. Then 8:00 to 8:15 is 15 minutes. 10 + 15 = 25 minutes.
e1_s2	Set 3: A tour of the balloon museum next to the field starts at 10:45 a.m. and lasts 50 minutes. What time does it end?	**11:35 a.m.** (also accepted: 11:35, 11:35 am, 11:35am, 11:35a.m., 11:35 a.m, 11:35 a. m.)	Jump from 10:45 to 11:00: 15 minutes. 50 − 15 = 35 minutes left. 11:00 + 35 minutes = 11:35 a.m.
e2_s2	Set 3: Two steps! Ms. Yazzie answered visitors' questions at the pilot booth from 11:40 a.m. to 1:05 p.m. How many minutes was that?	**85 minutes** (also accepted: 85, 85 minute, 85 min, 85 mins, 85min, 85mins, 1 hour 25 minutes, 1 hour and 25 minutes, 1 hour, 25 minutes, 1 hr 25 min, 1 h 25 min, 1 hr 25 mins)	11:40 a.m. to 12:00 noon is 20 minutes. 12:00 to 1:00 p.m. is 60 minutes. 1:00 to 1:05 is 5 minutes. 20 + 60 + 5 = 85 minutes (1 hour 25 minutes).

LIQUID VOLUME: LITERS OF WATER AND COCOA (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
v1_s0	Set 1: How many liters of water are in this measuring jug?	**7 liters** (also accepted: 7, 7 liter, 7 L, 7L)	The top of the water lines up with the mark between 6 and 8, which is 7 liters.
v2_s0	Set 1: Two steps! Nico has 5 jugs with 7 liters of water in each. He spills 3 liters on the way to the launch field. How many liters of water are left?	**32 liters** (also accepted: 32, 32 liter, 32 L, 32L)	Step 1: 5 jugs of 7 liters is 5 × 7 = 35 liters. Step 2: 35 − 3 = 32 liters.
v1_s1	Set 2: A big tank holds 64 liters of water. Ivy pours it equally into 8 jugs. How many liters go in each jug?	**8 liters** (also accepted: 8, 8 liter, 8 L, 8L)	Sharing 64 liters equally into 8 jugs means divide: 64 ÷ 8 = 8 liters in each jug.
v2_s1	Set 2: Which explanation is correct? Nico says a propane tank in a balloon basket holds about 2 liters. [He is wrong. 2 liters is about 2 large water bottles. A balloon propane tank holds much more, usually 40 liters or more. / He is right. A propane tank holds about 2 liters. / He is wrong, because a tank should be measured in minutes. / He is wrong. A propane tank holds less than 1 liter.]	**He is wrong. 2 liters is about 2 large water bottles. A balloon propane tank holds much more, usually 40 liters or more.**	A large water bottle holds about 1 liter, so 2 liters is only about 2 bottles of water. A balloon propane tank holds many more liters than that.

ID	Question	Answer	Why (explain to student)
v1_s2	Set 3: How many liters of water are in this big jug?	**13 liters** (also accepted: 13, 13 liter, 13 L, 13L)	Each small mark is 1 liter. The water is 3 marks above the 10, so it is 13 liters.
v2_s2	Set 3: Two steps! Ivy fills 3 coffee urns with 9 liters in each urn. Nico fills 1 big cocoa urn with 14 liters. How many liters did they fill in all?	**41 liters** (also accepted: 41, 41 liter, 41 L, 41L)	Step 1: Ivy fills 3 × 9 = 27 liters. Step 2: 27 + 14 = 41 liters in all.

MASS: GRAMS AND KILOGRAMS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
m1_s0	Set 1: Luz puts a box of Balloon Fiesta pins on the scale. What is its mass, in grams?	**350 grams** (also accepted: 350, 350 gram, 350 g, 350g)	The labels go up by 100, and each small mark is 50 grams. The pointer is one mark past 300: 300 + 50 = 350 grams.
m2_s0	Set 1: Find the mistake. Nico says a bag of green chile from a food stand has a mass of about 2 grams. What went wrong? [Grams are for very light things, like a paper clip. A bag of green chile is about 2 kilograms. / Nothing is wrong. A bag of green chile is about 2 grams. / He should say 2 liters, because the chile is in a bag. / A bag of green chile is about 2,000 kilograms.]	**Grams are for very light things, like a paper clip. A bag of green chile is about 2 kilograms.**	A gram is about the mass of one paper clip. A whole bag of green chile is much heavier, so kilograms is the right unit.
m1_s1	Set 2: Two steps! A wagon holds 6 sandbags. Each sandbag has a mass of 4 kilograms. The empty wagon has a mass of 3 kilograms. What is the total mass of the full wagon?	**27 kilograms** (also accepted: 27, 27 kilogram, 27 kg, 27 kgs, 27 kilos, 27kg, 27kgs, 27kilos)	Step 1: 6 sandbags × 4 kilograms = 24 kilograms. Step 2: 24 + 3 = 27 kilograms.
m2_s1	Set 2: A bag of Balloon Fiesta pins has a mass of 92 grams. After Luz trades some pins away, the bag has a mass of 58 grams. How many grams of pins did she trade away?	**34 grams** (also accepted: 34, 34 gram, 34 g, 34g)	The bag went from 92 grams to 58 grams. 92 − 58 = 34 grams traded away.
m1_s2	Set 3: Ivy splits 45 kilograms of sand into 5 equal sandbags. What is the mass of each sandbag, in kilograms?	**9 kilograms** (also accepted: 9, 9 kilogram, 9 kg, 9 kgs, 9 kilos, 9kg, 9kgs, 9kilos)	Sharing 45 kilograms into 5 equal sandbags means divide: 45 ÷ 5 = 9 kilograms.
m2_s2	Set 3: Which explanation is correct? Nico wants to know how much cocoa an urn holds. Which unit should he use? [Liters, because he wants to know how much liquid the urn holds. / Kilograms, because cocoa is a liquid. / Grams, because a cup of cocoa is small. / Minutes, because cocoa takes time to heat.]	**Liters, because he wants to know how much liquid the urn holds.**	Liters measure liquid volume, which is how much liquid a container holds. Grams and kilograms measure how heavy something is.

LAUNCH FIELD CHALLENGE: TWO-STEP PROBLEMS (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
c1_s0	Set 1: Two steps! Each cocoa urn takes 25 minutes to heat up. Ivy starts heating the first urn at 1:15 p.m. and starts the second urn the moment the first one is ready. What time is the second urn ready?	**2:05 p.m.** (also accepted: 2:05, 2:05 pm, 2:05pm, 2:05p.m., 2:05 p.m, 2:05 p. m.)	Step 1: 2 urns of 25 minutes is 25 + 25 = 50 minutes. Step 2: 1:15 + 45 minutes = 2:00, then 5 more minutes = 2:05 p.m.

ID	Question	Answer	Why (explain to student)
c2_s0	Set 1: Two steps! A water cooler has 20 liters of water. Nico fills 3 jugs with 4 liters each. He pours the rest into a bucket for washing up. How many liters go into the bucket?	**8 liters** (also accepted: 8, 8 liter, 8 L, 8L)	Step 1: 3 jugs × 4 liters = 12 liters. Step 2: 20 − 12 = 8 liters go into the bucket.
c1_s1	Set 2: Two steps! The crew's burrito breakfast starts at 8:40 a.m. Cooking takes 35 minutes, and then eating takes 20 minutes. What time does breakfast end?	**9:35 a.m.** (also accepted: 9:35, 9:35 am, 9:35am, 9:35a.m., 9:35 a.m, 9:35 a. m.)	Step 1: 35 + 20 = 55 minutes in all. Step 2: 8:40 + 20 minutes = 9:00, then 35 more minutes = 9:35 a.m.
c2_s1	Set 2: Two steps! A gas balloon pilot carries 3 bags of sand to use as ballast. Each bag has a mass of 8 kilograms. To rise higher, the pilot pours out 10 kilograms of sand. How many kilograms of sand are left?	**14 kilograms** (also accepted: 14, 14 kilogram, 14 kg, 14 kgs, 14 kilos, 14kg, 14kgs, 14kilos)	Step 1: 3 bags × 8 kilograms = 24 kilograms of sand. Step 2: 24 − 10 = 14 kilograms left.
c1_s2	Set 3: Two steps! A balloon drawing class at the kids' tent lasted 1 hour 15 minutes. It ended at 3:05 p.m. What time did it start?	**1:50 p.m.** (also accepted: 1:50, 1:50 pm, 1:50pm, 1:50p.m., 1:50 p.m, 1:50 p. m.)	Jump back 1 hour from 3:05 to 2:05. Then jump back 15 more minutes: 2:05 back 5 is 2:00, back 10 more is 1:50 p.m.
c2_s2	Set 3: Two steps! Luz has 90 grams of cocoa powder. She stirs 9 grams into each of 6 cups of hot milk. How many grams of cocoa powder are left?	**36 grams** (also accepted: 36, 36 gram, 36 g, 36g)	Step 1: 6 cups × 9 grams = 54 grams. Step 2: 90 − 54 = 36 grams left.

MATCH-UP ROUND (2 POINTS PER FIRST-TRY MATCH)

ID	Question	Answer	Why (explain to student)
match_s0_1	Set 1: Match elapsed time	**how much time passes from start to end**	2 points if matched on the first try.
match_s0_2	Set 1: Match liter	**a unit for liquid volume**	2 points if matched on the first try.
match_s0_3	Set 1: Match gram	**a unit of mass for light things**	2 points if matched on the first try.
match_s0_4	Set 1: Match kilogram	**a unit of mass for heavier things**	2 points if matched on the first try.
match_s0_5	Set 1: Match minute hand	**the long hand that shows the minutes**	2 points if matched on the first try.
match_s1_1	Set 2: Match 8:15 to 8:50	**35 minutes**	2 points if matched on the first try.
match_s1_2	Set 2: Match 9:40 to 10:10	**30 minutes**	2 points if matched on the first try.
match_s1_3	Set 2: Match Minute hand on the 9	**45 minutes after the hour**	2 points if matched on the first try.
match_s1_4	Set 2: Match 1 hour	**60 minutes**	2 points if matched on the first try.
match_s1_5	Set 2: Match 20 minutes after 2:50	**3:10**	2 points if matched on the first try.
match_s2_1	Set 3: Match 5 jugs of 6 liters	**30 liters**	2 points if matched on the first try.
match_s2_2	Set 3: Match A paper clip	**about 1 gram**	2 points if matched on the first try.
match_s2_3	Set 3: Match A large water bottle	**holds about 1 liter**	2 points if matched on the first try.
match_s2_4	Set 3: Match 4 sandbags of 7 kilograms	**28 kilograms**	2 points if matched on the first try.
match_s2_5	Set 3: Match 3 cocoa urns of 9 liters	**27 liters**	2 points if matched on the first try.

FIX THE SENTENCE (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
fs1_s0	Set 1: Fix the sentence: the balloons fly over albuquerque	**The balloons fly over Albuquerque.**	Start with a capital T. Albuquerque is the name of a city, so it gets a capital A. End the telling sentence with a period.
fs2_s0	Set 1: Fix the sentence: we filled jugs heated cocoa and loaded sandbags	**We filled jugs, heated cocoa, and loaded sandbags.** (also accepted: We filled jugs, heated cocoa and loaded sandbags.)	Start with a capital W, put commas between the things in the list, and end with a period.
fs1_s1	Set 2: Fix the sentence: what time does the first balloon launch	**What time does the first balloon launch?**	Start with a capital W. This sentence asks something, so it ends with a question mark.
fs2_s1	Set 2: Fix the sentence: ivy nico and luz carried the sandbags	**Ivy, Nico, and Luz carried the sandbags.** (also accepted: Ivy, Nico and Luz carried the sandbags.)	Each name gets a capital letter, commas go between the names in the list, and the sentence ends with a period.
fs1_s2	Set 3: Fix the sentence: how many liters of water did nico carry	**How many liters of water did Nico carry?**	Start with a capital H. Nico is a name, so it gets a capital N. The sentence asks something, so it ends with a question mark.
fs2_s2	Set 3: Fix the sentence: the two sandbags was full of sand	**The two sandbags were full of sand.**	Start with a capital T and end with a period. Two sandbags means more than one, so use were, not was.

BOSS BATTLE: THE BIG LAUNCH MORNING (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb1_s0	Set 1: The crew's lunch break starts at 11:50 a.m. and ends at 12:35 p.m. How many minutes long is the lunch break?	**45 minutes** (also accepted: 45, 45 minute, 45 min, 45 mins, 45min, 45mins)	11:50 a.m. to 12:00 noon is 10 minutes. 12:00 to 12:35 p.m. is 35 minutes. 10 + 35 = 45 minutes.
bb2_s0	Set 1: Which explanation is correct? Nico pours 4 liters of water into each of 7 jugs. Ivy says he used 11 liters of water. [Ivy is wrong. Nico filled 7 equal groups of 4 liters: 7 × 4 = 28 liters. / Ivy is right. 7 + 4 = 11 liters. / Ivy is wrong. It is 74 liters. / Ivy is wrong. It is 3 liters, because 7 − 4 = 3.]	**Ivy is wrong. Nico filled 7 equal groups of 4 liters: 7 × 4 = 28 liters.**	7 jugs with 4 liters in each are equal groups, so multiply: 7 × 4 = 28 liters.
bb1_s1	Set 2: Ivy watches the evening Balloon Glow from 6:48 p.m. to 7:29 p.m. How many minutes did she watch?	**41 minutes** (also accepted: 41, 41 minute, 41 min, 41 mins, 41min, 41mins)	6:48 to 7:00 is 12 minutes. 7:00 to 7:29 is 29 minutes. 12 + 29 = 41 minutes.
bb2_s1	Set 2: A water tank holds 72 liters. Nico fills it using a jug that holds 9 liters. How many full jugs does it take to fill the tank?	**8 jugs** (also accepted: 8, 8 jug, 8 full jugs, eight, eight jugs)	Each jug is a group of 9 liters. 72 ÷ 9 = 8, because 8 × 9 = 72. He needs 8 jugs.

ID	Question	Answer	Why (explain to student)
bb1_s2	Set 3: Pin trading at the pin tent runs from 9:30 a.m. to 11:05 a.m. How many minutes is that?	**95 minutes** (also accepted: 95, 95 minute, 95 min, 95 mins, 95min, 95mins, 1 hour 35 minutes, 1 hour and 35 minutes, 1 hour, 35 minutes, 1 hr 35 min, 1 h 35 min, 1 hr 35 mins)	9:30 to 10:00 is 30 minutes. 10:00 to 11:00 is 60 minutes. 11:00 to 11:05 is 5 minutes. 30 + 60 + 5 = 95 minutes (1 hour 35 minutes).
bb2_s2	Set 3: Which explanation is correct? A jug holds 4 liters of cocoa. A sandbag has a mass of 4 kilograms. Luz says they measure the same thing. [Luz is wrong. Liters measure how much liquid something holds. Kilograms measure how heavy something is. / Luz is right. 4 liters and 4 kilograms are the same thing. / Luz is wrong, because liters are only for time. / Luz is right, because both numbers are 4.]	**Luz is wrong. Liters measure how much liquid something holds. Kilograms measure how heavy something is.**	Liters measure liquid volume. Kilograms measure mass. They answer different questions, so they are not the same.

BOSS BATTLE: THE LAST LANDING (5 POINTS EACH)

ID	Question	Answer	Why (explain to student)
bb3_s0	Set 1: Two small baskets of Balloon Fiesta pins have masses of 245 grams and 380 grams. What is their total mass, in grams?	**625 grams** (also accepted: 625, 625 gram, 625 g, 625g)	Add the two masses, both in grams: 245 + 380 = 625 grams.
bb4_s0	Set 1: Find the mistake. Luz says 40 minutes after 10:35 a.m. is 10:75 a.m. Type the correct time.	**11:15 a.m.** (also accepted: 11:15, 11:15 am, 11:15am, 11:15a.m., 11:15 a.m, 11:15 a. m.)	There is no such time as 10:75, because an hour has only 60 minutes. 10:35 + 25 minutes = 11:00, then 15 more minutes = 11:15 a.m.
bb3_s1	Set 2: Find the mistake. Ivy reads this clock as 2:11. What went wrong? [The minute hand has passed the 11, so count by fives to 55, then 1 more: it is 2:56. / Nothing is wrong. The minute hand is near the 11, so it is 2:11. / It is 3:56, because the hour hand is close to the 3. / It is 2:51, because the minute hand is 1 mark past the 10.]	**The minute hand has passed the 11, so count by fives to 55, then 1 more: it is 2:56.**	The minute hand counts minutes, not hours. On the 11 it means 55 minutes. One more mark is 56. The hour hand has not reached 3, so it is 2:56.
bb4_s1	Set 2: Two steps! Ms. Yazzie's chase truck carries 4 sandbags with a mass of 9 kilograms each, and 1 basket of snacks with a mass of 7 kilograms. What is the total mass?	**43 kilograms** (also accepted: 43, 43 kilogram, 43 kg, 43 kgs, 43 kilos, 43kg, 43kgs, 43kilos)	Step 1: 4 sandbags × 9 kilograms = 36 kilograms. Step 2: 36 + 7 = 43 kilograms.
bb3_s2	Set 3: Two steps! Nico had 50 liters of water. He filled 3 coolers with 8 liters in each cooler. How many liters of water are left?	**26 liters** (also accepted: 26, 26 liter, 26 L, 26L)	Step 1: 3 coolers × 8 liters = 24 liters. Step 2: 50 − 24 = 26 liters left.
bb4_s2	Set 3: Find the mistake. Ivy's cocoa cooled for 25 minutes and was ready at 5:10 p.m. She says it started cooling at 5:35 p.m. Type the correct time it started cooling.	**4:45 p.m.** (also accepted: 4:45, 4:45 pm, 4:45pm, 4:45p.m., 4:45 p.m, 4:45 p. m.)	To find a start time, jump backward. 5:10 back 10 minutes is 5:00. Back 15 more minutes is 4:45 p.m. Ivy jumped forward instead of backward.

Pre & Post sheet answer guide

Many items are open-ended. Grade for genuine understanding and complete sentences, not exact wording. Sheet: grade3/prepost/KCE_Math_PrePost_G3W19.pdf.

PRE-LESSON WARM-UP

#	Prompt	Model answer / grading note
1	Review: compare 2/3 and 2/6	2/3 > 2/6. Same numerator, and thirds are bigger pieces than sixths.
2	Draw a clock showing 9:15	Hour hand just past the 9; minute hand on the 3.
3	Lifts off at 7:00 a.m., flies 1 hour	8:00 a.m.
4	Measured in liters	water in a cooler
5	More mass: 1 gram or 1 kilogram	1 kilogram

POST-LESSON CHALLENGE

#	Prompt	Model answer / grading note
1	Hour hand between 3 and 4; minute hand 2 marks past the 6	3:32 (count 5, 10, ... 30, then 31, 32).
2	Pin table open 10:35 a.m. to 11:20 a.m.	45 minutes: 10:35 to 11:00 is 25 minutes, and 11:00 to 11:20 is 20 minutes.
3	4 jugs of 6 liters, then 5 liters poured out	Step 1: 4 × 6 = 24 liters. Step 2: 24 − 5 = 19 liters left.
4	Ivy says 3 + 7 + 6 = 16 kilograms	She added the 3 instead of multiplying. 3 sandbags × 7 kilograms = 21 kilograms, and 21 + 6 = 27 kilograms.

★ CHALLENGE

#	Prompt	Model answer / grading note
1	Balloon Glow at 6:15 p.m.; setup takes 40 minutes	The crew must start at 5:35 p.m. (count back 15 minutes to 6:00, then 25 more minutes to 5:35).